Unit 3 Review: Distributions — ANSWER KEY

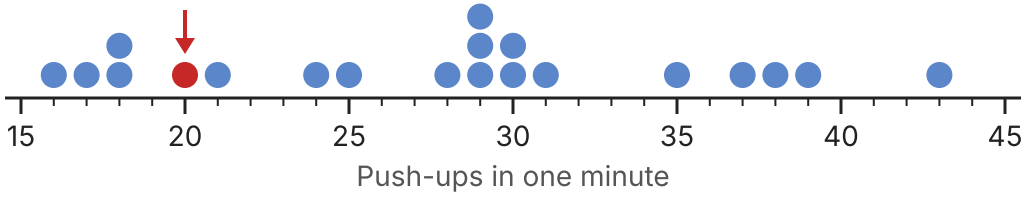
3.1 Percentiles

1. 9.9 is at the **46.15**th percentile — 46 or 47 (rounded down or up) is fine

Values below 9.9: **5.3**, **6.1**, **8.0**, **8.0**, **8.7**, **9.7**, **[9.9]**, 10.7, 10.8, 11.3, 12.1, 12.3, 13.1 → **6** of the 13 values.
6 ÷ 13 ≈ 0.4615 → 46.15%
9.9 is at about the **46th percentile** (rounding down; 47th is fine too): about 46.2% of the data values are less than 9.9.

2. Push-ups = **20** push-ups — answers that round to this are fine

We want 20% of 20 = 4 values below.
Counting from the left: 1 at 16, 1 at 17, 2 at 18 → that is 4 dots, so the next value, **20**, has exactly 4 values below it.
Check: **4 ÷ 20 = 20%**. So the 20th percentile is **20**.
The 20th-percentile value, marked in red:



3. **A. 37% of students in the grade have less daily screen time than Amara.**

The 37th percentile means 37% of the values in the group are less than this one.
So the correct statement is: **37% of students in the grade have less daily screen time than Amara.**

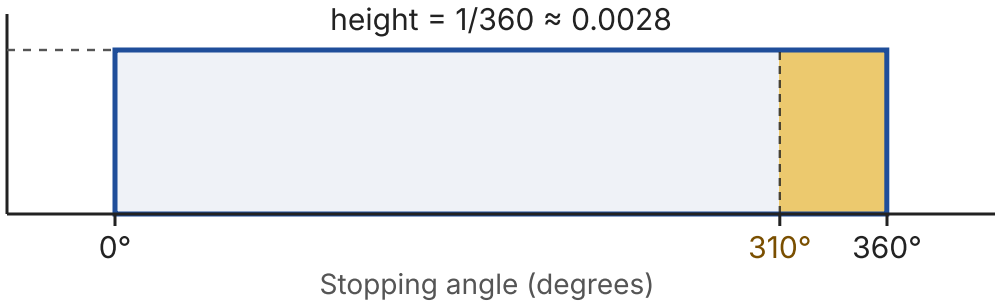
4. **B. It goes down.**

Before: **5 ÷ 25 = 20%**. After: **5 ÷ 26 ≈ 19.23%**.
The same number of students are below, but the class is bigger. So the percentile **goes down**.

3.2 Uniform Distributions

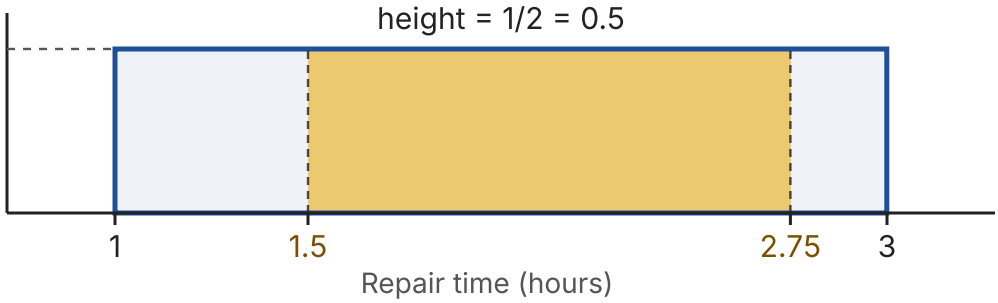
5. Percent = **13.89** % — 13 or 14 (rounded down or up) is fine

Density curve: a rectangle from 0° to 360°, height = **1 ÷ 360 = 1/360 ≈ 0.0028**.
Width of the shaded region: **360 − 310 = 50**
Area: **50 ÷ 360 ≈ 0.1389** → ≈ **13.89%**



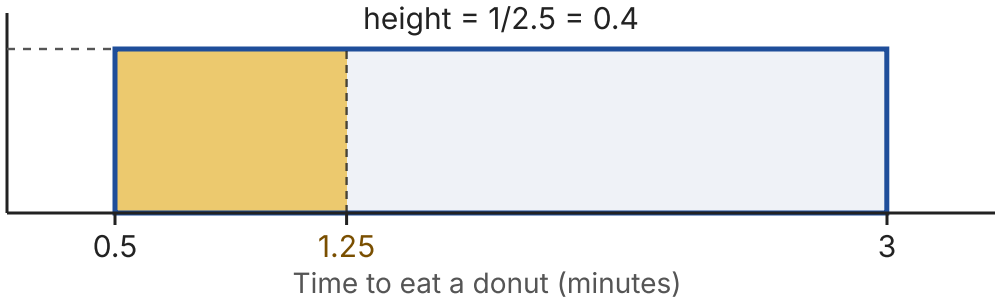
6. Percent = **62.5** % — 62 or 63 (rounded down or up) is fine

Density curve: a rectangle from 1 to 3, height = **1 ÷ 2 = 1/2 = 0.5**.
Width of the shaded region: **2.75 − 1.5 = 1.25**
Area: **1.25 ÷ 2 = 0.625** → **62.5%**



7. 30th percentile = **1.25** minutes — answers that round to this are fine

Density curve: a rectangle from 0.5 to 3, height = **1 ÷ 2.5 = 1/2.5 = 0.4**.
We need 30% of the area to the left: width = **0.3 × 2.5 = 0.75**
Start at a = 0.5: **0.5 + 0.75 = 1.25**, so the 30th percentile is **1.25** minutes.



8. **A. From 30 to 60, height ≈ 0.03333**

Width = 30, so height = **1 ÷ 30 ≈ 0.0333** (then width × height = 1).
Correct: **From 30 to 60, height ≈ 0.03333**

3.3 z-Scores

9. z = **−1.11** — answers that round to this are fine

z = (1100 − 1200) ÷ 90 = −100 ÷ 90 ≈ −1.11

So the value is 1.11 standard deviations below the mean.

10. **A. Elijah**

Elijah: **z = (38.4 − 51) ÷ 5.4 = −12.6 ÷ 5.4 ≈ −2.33**

Marcus: **z = (21.4 − 40.5) ÷ 9.2 = −19.1 ÷ 9.2 ≈ −2.08**

Here **lower is better**, so the **smaller** (more negative) z-score wins. −2.33 vs. −2.08 → **Elijah** did better relative to their group.

11. Wingspan = **9.64** meters — answers that round to this are fine

x = μ + z·σ = 14.5 + (−2.7)(1.8)

= 14.5 − 4.86 = 9.64 → 9.64 meters

12. **C. 2.64 pounds below the mean**

2.2 standard deviations × 1.2 pounds each = 2.64 pounds.

Negative/positive z → below the mean. Correct: **2.64 pounds below the mean**

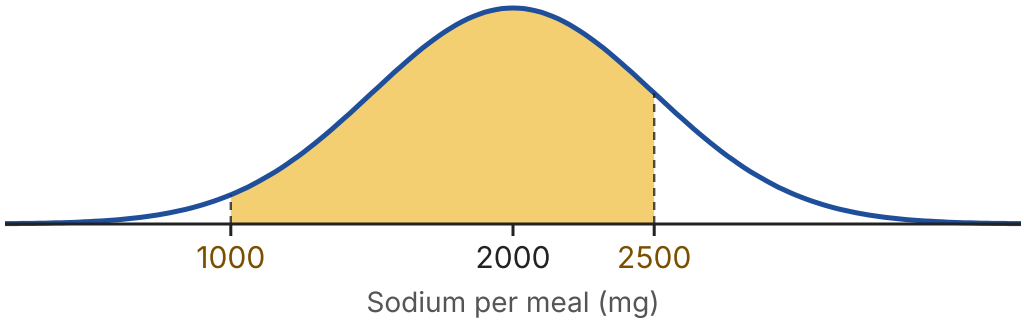
3.4 Normal Distributions (Desmos)

13. Percent = **81.86** % — 81 or 82 (rounded down or up) is fine

Desmos: **normaldist(2000, 500)**, cumulative probability from 1000 to 2500 → 0.8186

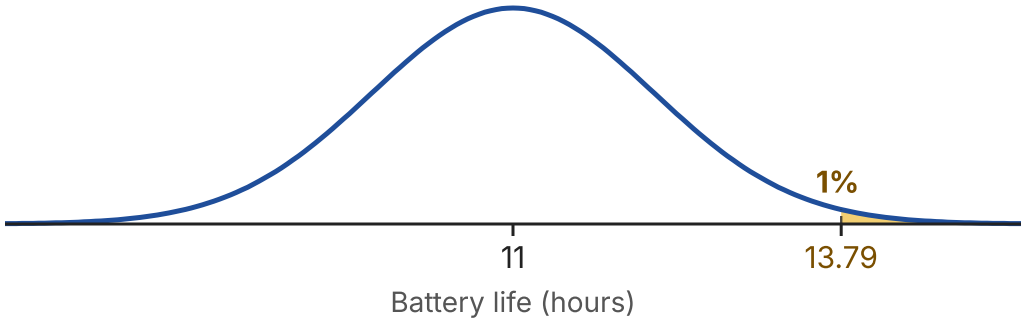
So about **81.86%** of meals have between 1000 and 2500 mg of sodium.

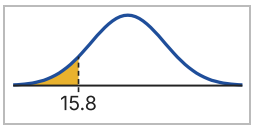
Shaded region:



14. Cutoff = **13.79** hours — answers that round to this are fine

Top 1% → left area = 1 − 0.01 = 0.99: **normaldist(11, 1.2).inversecdf(0.99)**
Cutoff ≈ **13.79 hours**
Shaded region:



15. **C** 

"weigh less than 15.8 ounces" means everything to the **left** of 15.8.

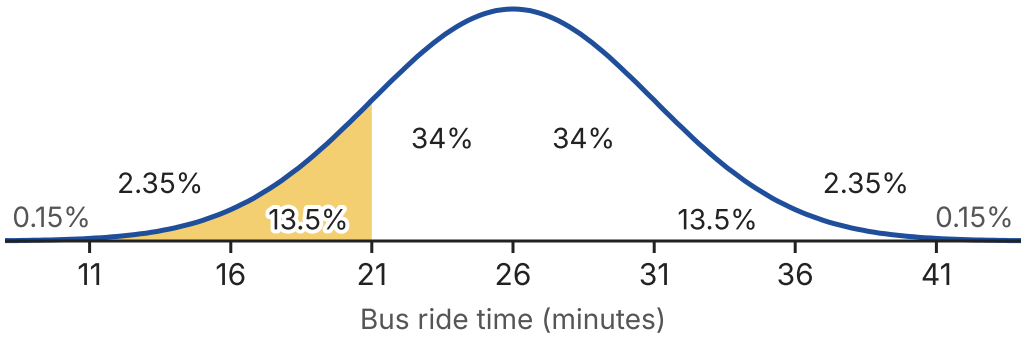
16. **D. 45% of cars get less than 23.25 mpg.**

The 45th percentile means 45% of the values are less than it.
Correct: **45% of cars get less than 23.25 mpg.**

3.5 The Empirical Rule

17. Percent = **16** % — answers that round to this are fine

21 = 26 − 5, which is **1 SD below the mean**.
Area below it: **0.15 + 2.35 + 13.5 = 16%**
Shaded region:



18. Delivery time = **18** minutes — answers that round to this are fine

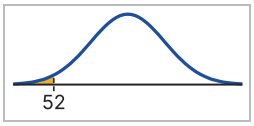
Area to the left of $\mu - 2\sigma$ is **0.15 + 2.35 = 2.5%**.
So the 2.5th percentile is 2 SDs below the mean: **30 − 2 × 6 = 18 minutes**
Shaded region:



19. **B. 2.5%**

95% are within 2 standard deviations, so 5% are outside, split evenly between the two tails: 2.5% below.
So about **2.5%** of students have an SAT math score more than 2 standard deviations below the mean.

20. **C**



"resting heart rate is more than 2 standard deviations below the mean" means everything to the **left** of 52.